

```
-- Creating new database for Task 06-----
create database Online_sales_Task06 ;

---- using Online_sales_Task06-----
use Online_sales_Task06;

--Show table online_sales_Task96-----

select * from dbo.Online_Sales_Data;

---- count Total transaction -----
select count(*) as "Total_Transaction"
from Online_Sales_Data;

---Total Revenue -----
select sum(total_revenue )as "Total Revenue" from Online_Sales_Data;

-----Top 10 products by revenue-----
select top 10 Product_name,sum(total_revenue) as "Total_Revenue" from
Online_Sales_Data group by Product_Name order by "Total_Revenue" desc;

-----Revenue by catergory-----
select product_category, sum(Total_revenue) as "Total_Revenue"
from Online_Sales_Data group by Product_Category order by Total_Revenue desc;

-----Revenue by region-----
select region,sum(total_revenue) as "Total_Revenue"
from online_sales_data group by Region order by "Total_Revenue" desc;

-----Unit sole by region-----
select Region, sum(Units_Sold) as "Sold_Unit" from Online_Sales_Data
group by region order by "Sold_Unit" desc;

-----Most used paymets methods-----
select payment_Method,sum(total_revenue) as "Payments" from Online_Sales_Data
group by Payment_Method order by "Payments" desc;

----- Revenue by payment method-----
select Payment_method, sum(Total_revenue) as "total_Revenue" from Online_Sales_Data
group by payment_method order by "total_Revenue" desc;

----- Time analysis , monthly revenue -----
select month(date) as "Month", year(date) as "Year", sum(Total_revenue) as
"Total_revenue"
from Online_Sales_Data group by month(date), year(date) order by "Total_revenue"
desc;

----- yearly Revenue-----
select Year(date) as "Year", sum(Total_Revenue) as "Total_Revenue" from
online_sales_data
group by Year(date) order by "total_Revenue" desc;
```

----Which product generated the highest revenue? ----

```
select top 10 Product_Name, sum(Total_revenue) as "Total_Revenue" from
Online_Sales_Data
group by Product_Name order by "Total_Revenue" desc;
```

---Which category sold the most units?---

```
select Product_Category, sum(Total_revenue) as "Total_Revenue" from Online_Sales_Data
group by Product_Category order by "Total_Revenue" desc;
```

----Which region generated the highest revenue?----

```
select Region, sum(Total_revenue) as "Total_Revenue" from Online_Sales_Data
group by Region order by "Total_Revenue";
```

-----Total_Revenue-----

```
select ROUND(sum(total_Revenue),2) as "Total_Revenue" from Online_Sales_Data;
```

-----Total Unit sold-----

```
select sum(Units_Sold) as "Unit_sold" from Online_Sales_Data;
```

-----Average Order Value-----

```
select round(avg(Total_revenue),2) as "Avg_order_Value" from online_sales_data;
```

-----Top 5 Products by Revenue-----

```
select top 5 Product_name, round(sum(Total_revenue),2) as "Product_Total_Revenue"
from Online_Sales_Data group by Product_Name order by Product_Total_Revenue desc;
```

----Top Category by revenue----

```
select product_category, round(sum(total_revenue),2) as "Total_Revenue"
from Online_Sales_Data group by Product_Category order by "Total_Revenue" desc;
```

----Revenue by Payment methods----

```
select Payment_Method, Round(sum(Total_revenue),2) as "Total_Revenue"
from Online_Sales_Data group by Payment_Method order by "Total_Revenue" desc;
```

----Monthly sales Trends----

```
select Year(date) as "Year", month(date) as "Month", round(sum(Total_revenue),2)
as "Total_Revenue" from Online_Sales_Data group by Year(date), month(date)
order by "total_Revenue" desc;
```

----Best Sales Months-----

```
select Month(date) as "Month", Year(date) as "Year", round(sum(Total_revenue),2)
as "Total_revenue" from online_sales_data group by Month(date), Year(date)
order by "Total_Revenue" desc;
```

-----Highest Revenue Product in Each Category-----

```
SELECT product_category,
product_name,
"revenue"
```

```
FROM (
SELECT product_category,
       product_name,
       SUM(Total_Revenue) AS "revenue",
       RANK() OVER(
         PARTITION BY product_category
         ORDER BY SUM(total_revenue) DESC) rnk
FROM Online_Sales_Data
GROUP BY product_category,
         product_name
) t
WHERE rnk=1;
```

-----Running Revenue Total-----

```
select Date, Total_Revenue, sum(Total_revenue) over(order by date) as "Runing Total"
from online_sales_data;
```

-----Top 3 Products in Each Region-----

```
SELECT *
FROM
(
SELECT region,
       product_name,
       SUM(total_revenue) as "revenue",
       DENSE_RANK() OVER
         (PARTITION BY region
          ORDER BY SUM(total_revenue) DESC) rnk
FROM Online_Sales_Data
GROUP BY region,
         product_name
) x
WHERE rnk <=3;
```

-----Most Popular Payment Method-----

```
SELECT top 3 payment_method,
       COUNT(*) as "total_transactions"
FROM Online_Sales_Data
GROUP BY payment_method
ORDER BY total_transactions DESC;
```